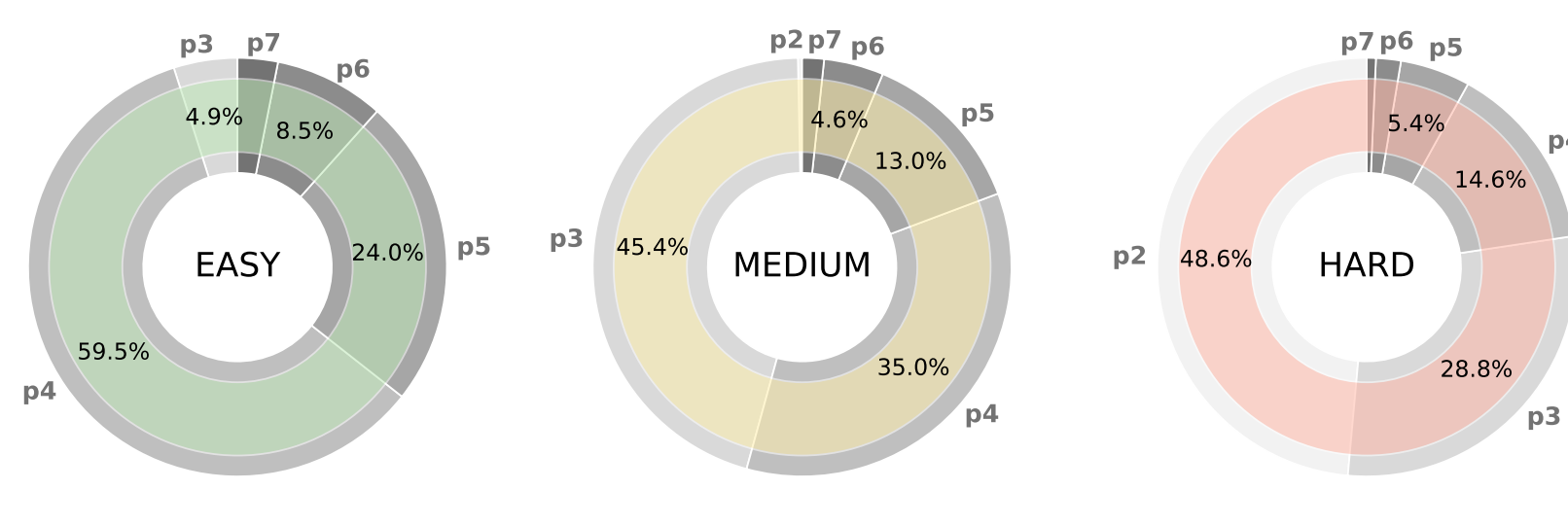
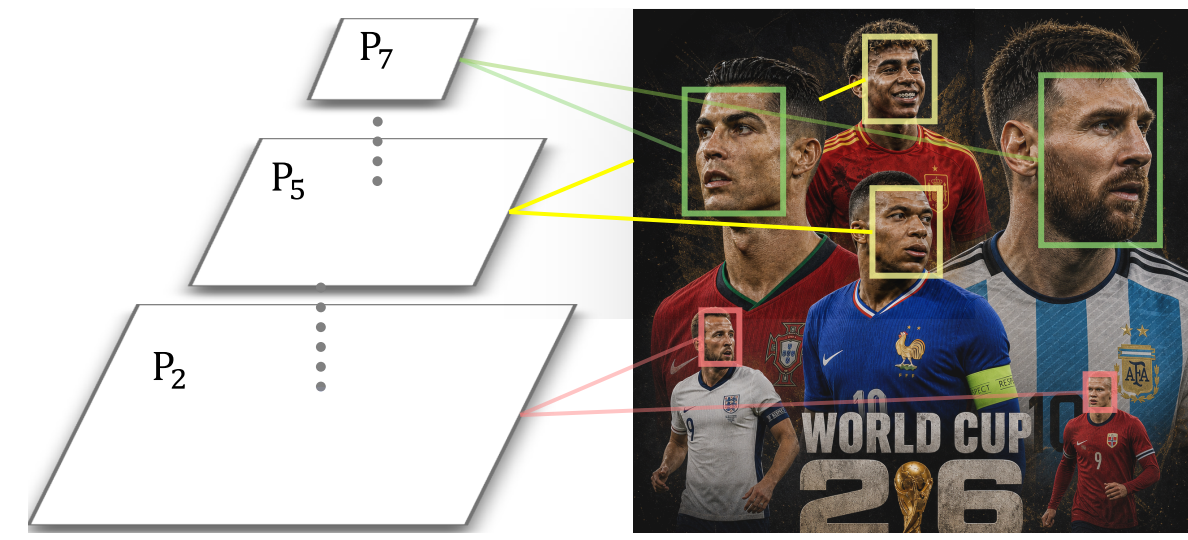


SelfX^TFace: Attention Based Feature Pyramid Network in Face Detection

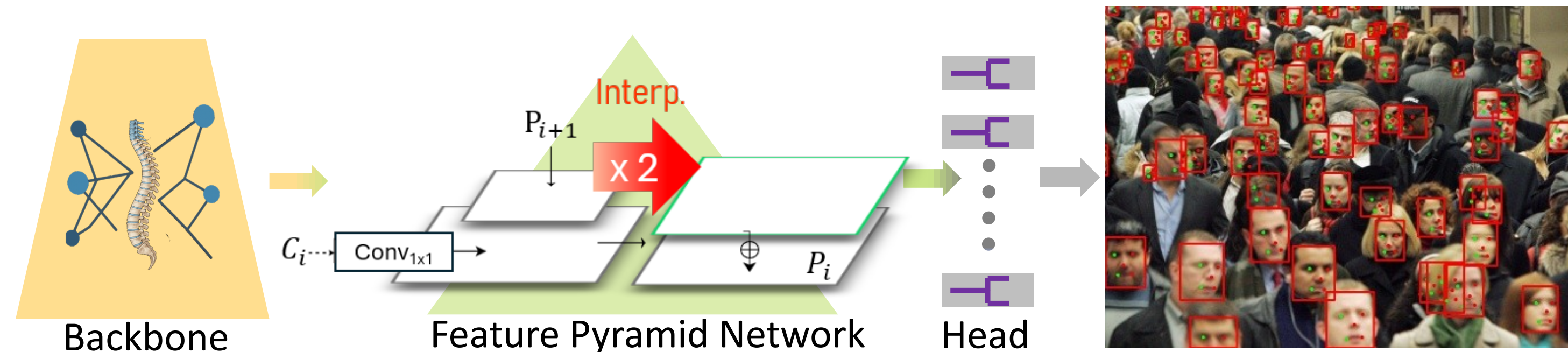
Furkan Kasım, Carlos Kirchdorfer, Salman Mohammad, Manuel Günther

INTERNATIONAL CONFERENCE ON PATTERN RECOGNITION, ICPR 2026

1. Introduction & Background

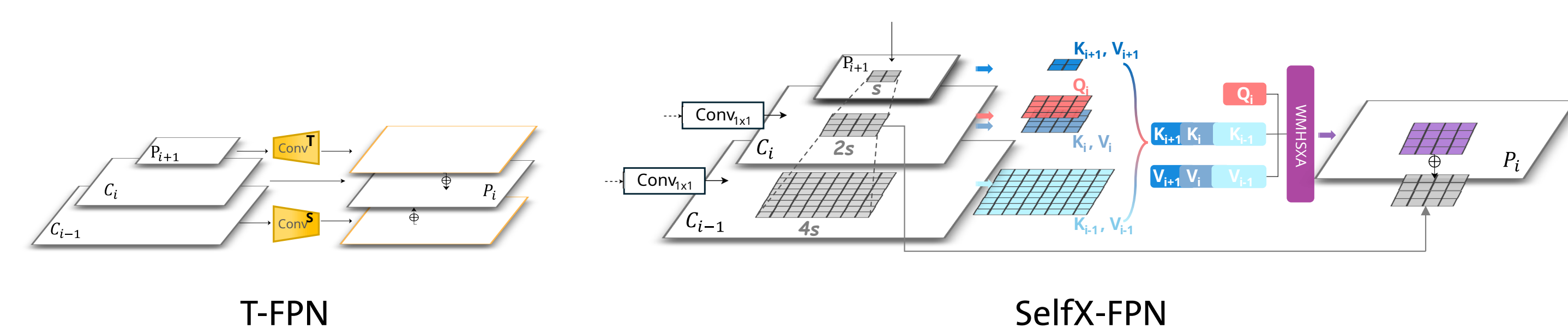


- Face detection suffers from **extreme scale variation**: $\sim 55\%$ of WIDER FACE [10] instances are smaller than 20px, far worse than objects in MS-COCO [4] ($\sim 18\%$).
- Modern face detectors follow a backbone–neck–head design, typically using a residual backbone and a **Feature Pyramid Network (FPN)** [3] neck to build a multi-scale feature representation.
- Standard FPNs **upsample feature maps via interpolation** and fuse them with shallow features through **element-wise summation**. This estimates pixel values without regard to semantic content, introducing **aliasing artifacts** especially harmful for small faces.

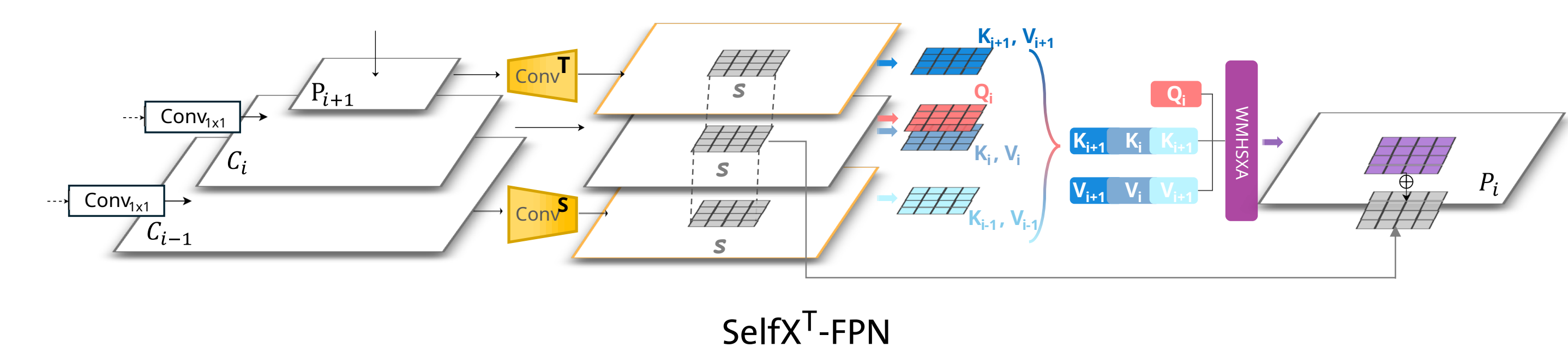


- We replace interpolation with **learnable transposed convolutions** and introduce **window-based self-cross attention** for scale fusion.
- SelfX^T-FPN** is a **lightweight, parameter-shared** FPN with consistent **gains** over prior FPN variants.

2. Approach



- Approach 1: T-FPN** replaces feature map interpolation with a learnable transposed convolution Conv^T for upsampling and a strided convolution Conv^S for downsampling, fused with learnable weights.
- Approach 2: SelfX-FPN** fuses adjacent pyramid levels via windowed multi-head attention **WMHSA**, each region adaptively aggregates fine- and coarse-scale context of feature maps with different window scales $s/2s/4s$ — **no feature map scaling needed**.



- Joint Approach: SelfX^T-FPN** first aligns resolutions via transposed/strided convolutions, then fused with WMHSA using a fixed s across levels, **unifying both mechanisms into one lightweight module**.

3. Results

- Comparing approaches over baselines and related work on ResNet-50 (**Table 1**):
 - Removing the FPN** entirely causes a **substantial performance drop**, especially on **Hard**.
 - T-FPN** alone improves AP across all subsets, **+1.3% on Hard**.
 - SelfX-FPN** reaches **+1.9% AP** on Hard
 - SelfX^T-FPN** pushes this to nearly **+2% AP**.
 - Both SelfX-FPN and SelfX^T-FPN **outperform all competing FPNs** while using **half the parameters** of the strongest competitor.

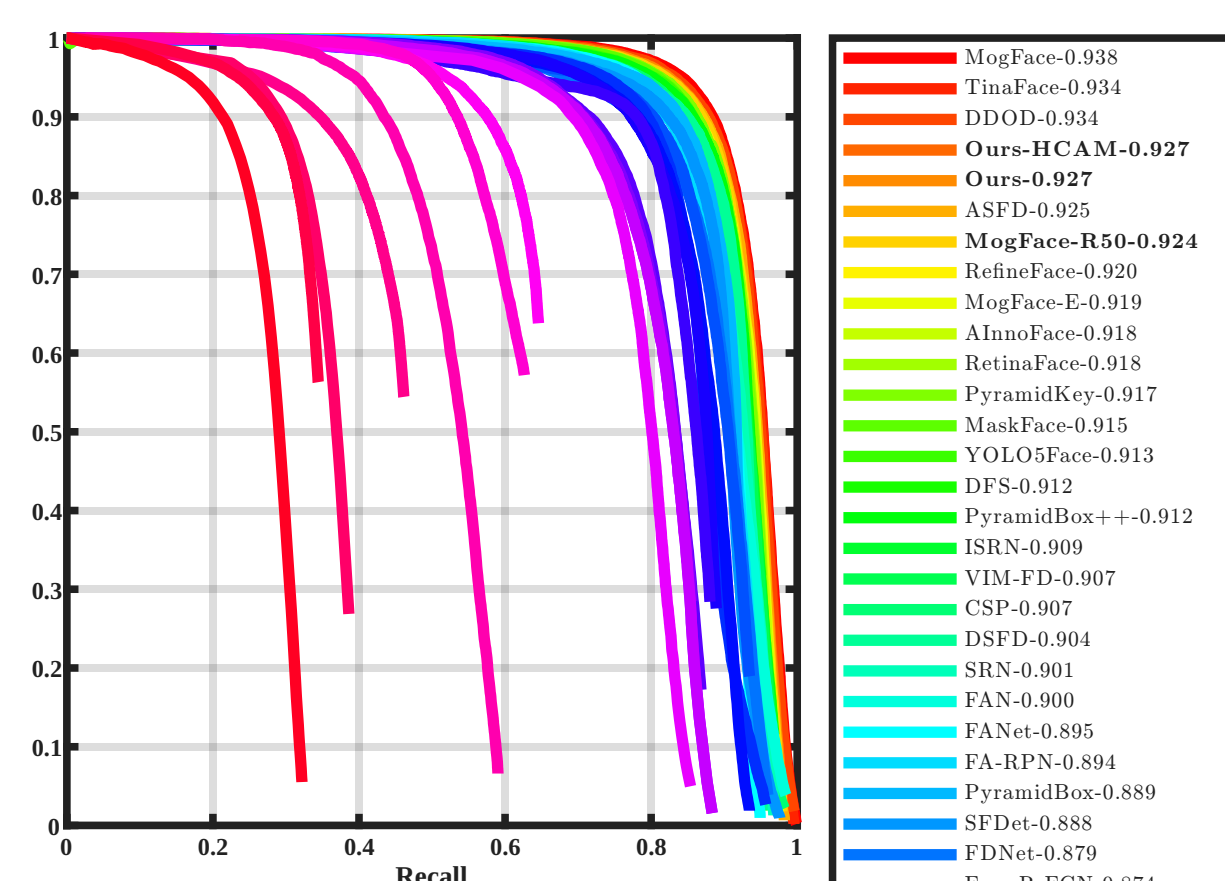
Method	Params	Easy	Medium	Hard
w/o FPN	4.6M	95.14	93.17	82.62
LFPN [8] (baseline)	8.7M	95.18	94.54	87.33
BiFPN [7]	1.8M	94.95	94.36	88.65
FPN [3]	4.6M	95.28	94.36	87.18
FEM [2]	6.9M	95.20	94.18	88.03
SepFPN [1]	7.3M	95.34	94.37	87.29
AFPN [9]	11.9M	95.48	94.66	89.03
PAFPN [5]	14.1M	95.51	94.60	88.68
<i>T-FPN</i> ③ (ours)	5.2M	95.39	94.65	88.68
<i>SelfX-FPN</i> ⑮ (ours)	4.8M	95.73	94.95	89.22
<i>SelfX^T-FPN</i> ⑳ (ours)	6M	95.80	95.08	89.30

Table 1: Comparison to Other FPNs

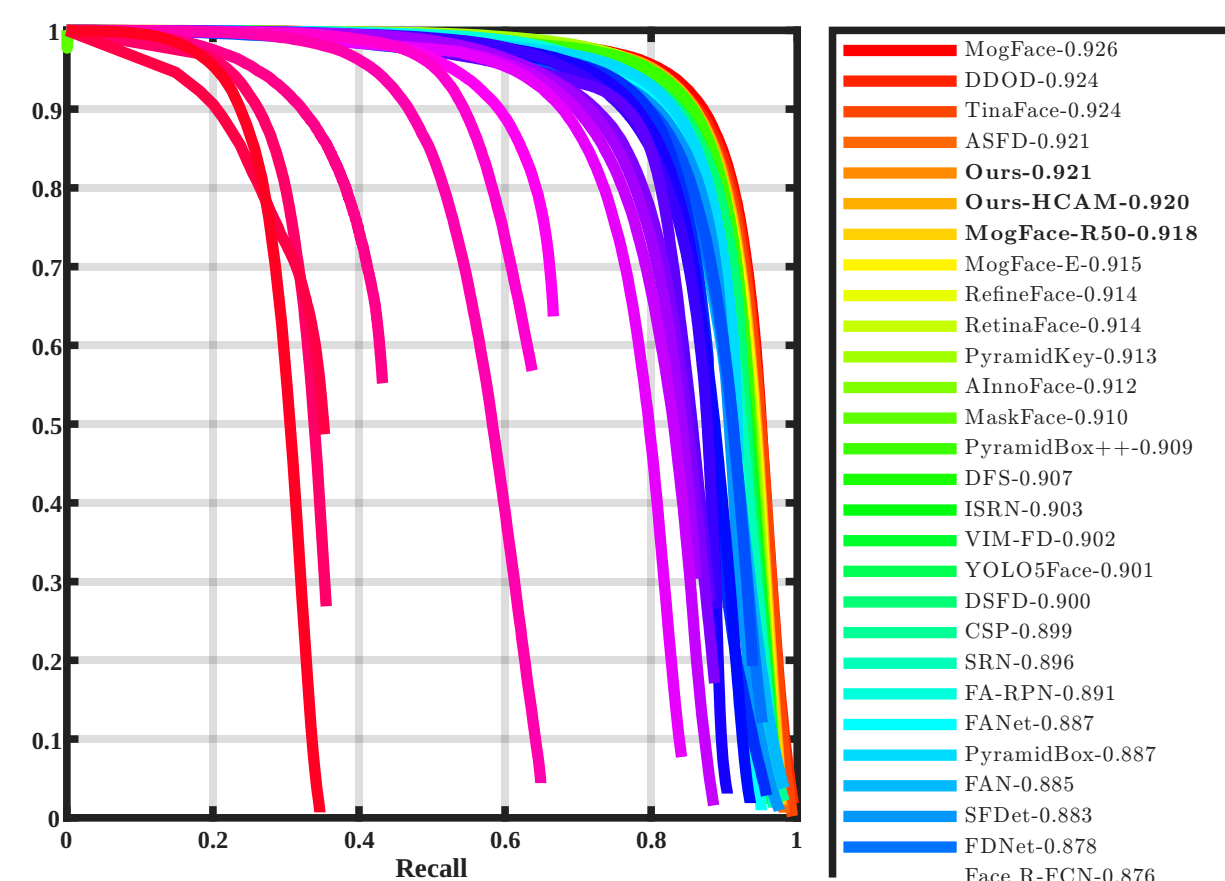
Model	Neck	HCAM	Easy	Medium	Hard	Params	ms
MogFace [6]	LFPN [8]	✓	95.53	94.78	90.84	41M	51
SelfX ^T Face	SelfX ^T (s=1)	✓	95.65	95.05	91.20	38M	65
SelfX ^T Face	SelfX ^T (s=2)	✓	95.61	95.02	91.40	38M	60
SelfX ^T Face	SelfX ^T (s=4)	✓	95.59	94.92	91.34	38M	61
SelfX ^T Face	SelfX ^T (s=2)	✗	95.65	95.01	91.15	38M	60

Table 2: Comparison to strongest SOTA Face Detector

- Comparing **SelfX^T-Face** over **MogFace [6]** (**Table 2**) and other **SOTA** detectors (**Precision-Recall curves**):
 - Integrating SelfX^T-FPN into a strong state-of-the-art detector **improves performance of all subsets**, with up to **0.6% AP gain** on Hard.
 - This gain comes at a **negligible efficiency cost** — only 18 GMACs and 9ms additional runtime.
 - Under **multi-scale testing**, **SelfX^T-Face** achieves **highly competitive Hard-subset performance** on both validation and test sets.
- The ablation study (**Table 3**) shows that self- and cross-scale **attention**, **weight sharing**, and **bidirectional** lower-level fusion **consistently improve performance**. **Relative positional encoding (RPE)** benefits *SelfX-FPN* but not *SelfX^T-FPN*, where Conv^T and Conv^S already align spatial features across scales.



Precision-Recall curves: Validation-Hard



Precision-Recall curves: Test-Hard

Index	Method	Up	self-X	s	Shared	Bi-dir	RPE	Easy	Medium	Hard
①	Baseline	Interpol.						95.18	94.54	87.33
②	<i>T-FPN</i>	Conv^T						95.38	94.76	88.68
③		Conv^T	✓					95.39	94.65	88.68
④		Conv^T				✓		95.55	94.67	88.81
⑤	<i>SelfX-FPN</i>	—	2					95.65	94.68	88.25
⑥		—	✓	2				95.57	94.72	88.79
⑦		—	✓	2	✓			95.87	94.92	88.85
⑧		—	✓	2		✓		95.52	94.68	88.84
⑨		—	✓	2	✓	✓		95.62	94.82	88.81
⑩		—	✓	2	✓	✓	✓	95.68	94.92	89.00
⑪		—	✓	1				95.72	94.82	89.07
⑫		—	✓	1	✓			95.51	94.75	88.90
⑬		—	✓	1		✓		95.36	94.53	89.08
⑭		—	✓	1	✓	✓		95.60	94.80	88.86
⑮		—	✓	1	✓	✓	✓	95.73	94.95	89.22
⑯	<i>SelfX^T-FPN</i>	Conv^T	✓	4	✓	✓	✓	95.73	94.91	88.68
⑰		Conv^T	✓	4	✓	✓	✓	95.67	94.89	88.53
⑱		Conv^T	✓	2	✓	✓	✓	95.66	94.77	89.21
⑲		Conv^T	✓	2	✓	✓	✓	95.50	94.78	89.24
⑳		Conv^T	✓	1	✓	✓	✓	95.80	95.08	89.30
㉑		Conv^T	✓	1	✓	✓	✓	95.59	94.85	89.24
㉒		Conv^T	✓	1	✓	✓	✓	95.96	95.14	89.03

Table 3: Ablation Study

4. Conclusions

- We close the gap between **attention-based fusion** used in object detectors and **upsampling-based FPNs** for face detection by employing windowed attention over self-, upper-, and lower-level features aligned via transposed/strided convolutions.
- Our approach **outperforms** standard and advanced FPNs, especially for **small-scale faces**.
- Sharing parameters** across pyramid levels dramatically **reduces parameter count** without hurting detection quality.
- Our FPN is **complementary** to other detection improvements, such as better loss functions or head designs.

References

- Joonhyun Jeong, Beomyoung Kim, Joonsang Yu, and Young Joon Yoo. EResFD: Rediscovery of the effectiveness of standard convolution for lightweight face detection. In WACV, 2022.
- Jian Li, Yabiao Wang, Changan Wang, Ying Tai, Jianjun Qian, et al. DSFD: Dual shot face detector. In CVPR, 2019.
- Tsung-Yi Lin et al. Feature pyramid networks for object detection. In CVPR, 2017.
- Tsung-Yi Lin, Michael Maire, Serge Belongie, James Hays, et al. Microsoft COCO: Common objects in context. In ECCV, 2014.
- Shu Liu, Lu Qi, Haifang Qin, Jianping Shi, and Jiaya Jia. Path aggregation network for instance segmentation. In CVPR, 2018.
- Yang Liu, Fei Wang, Jiankang Deng, Zhipeng Zhou, Baigui Sun, and Hao Li. MogFace: Towards a deeper appreciation on face detection. In CVPR, 2022.
- Mingxing Tan, Ruoming Pang, and Quoc V. Le. EfficientDet: Scalable and efficient object detection. In CVPR, 2019.
- Xu Tang, Daniel K. Du, Zeqiang He, and Jingtuo Liu. PyramidBox: A context-assisted single shot face detector. In ECCV, 2018.
- Guoyu Yang, Jie Lei, Hao Tian, Zunlei Feng, and Ronghua Liang. Asymptotic feature pyramid network for labeling pixels and regions. Transactions on Circuits and Systems for Video Technology (TCSVT), 2024.
- Shuo Yang, Ping Liu, Chen-Chang Loy, and Xiaoou Tang. WIDER FACE: A face detection benchmark. In CVPR, 2016.



github.com/AIML-IfI/SelfXtFace